Welcome to Get on My Level - Mastering LOD Calculations!

Open the **TC26-GetonMyLevel-Starter.twbx** file to do the challenges!

Intro: Multiple Aggregations

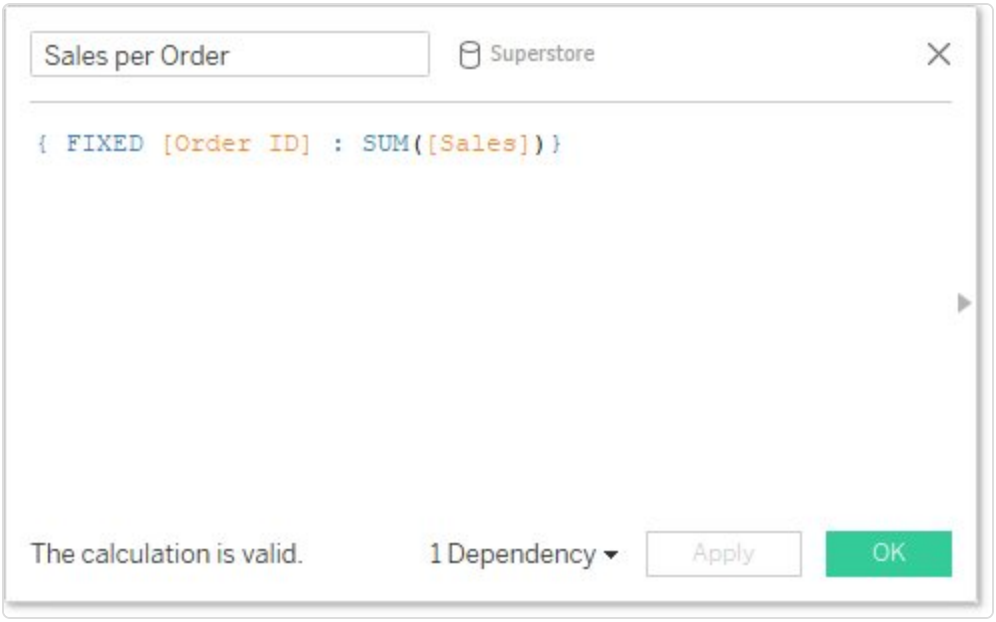
Exercise

What is the Average Sales per Order?

458.6

1. Write a calculation to get the Sum of Sales for each Order ID.

   Calculation:

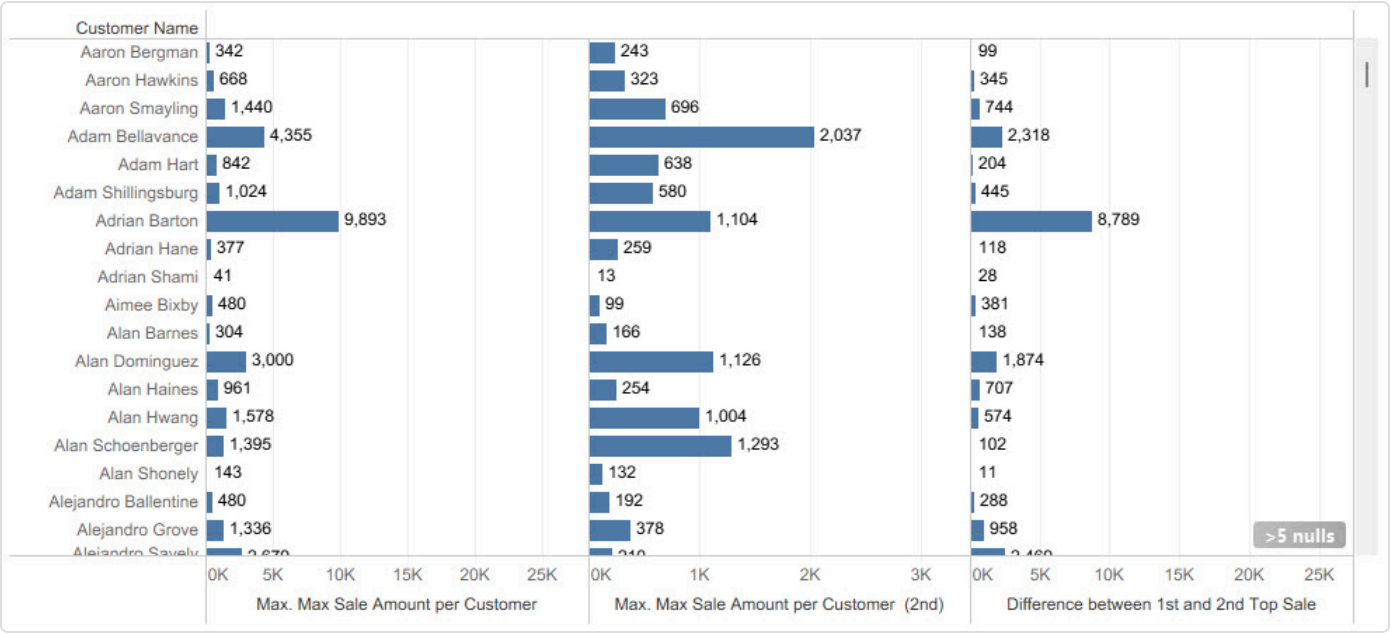


2. Drag the Sales per Order field to Rows, and change the aggregation type to `AVG()`.

1: Pivoting

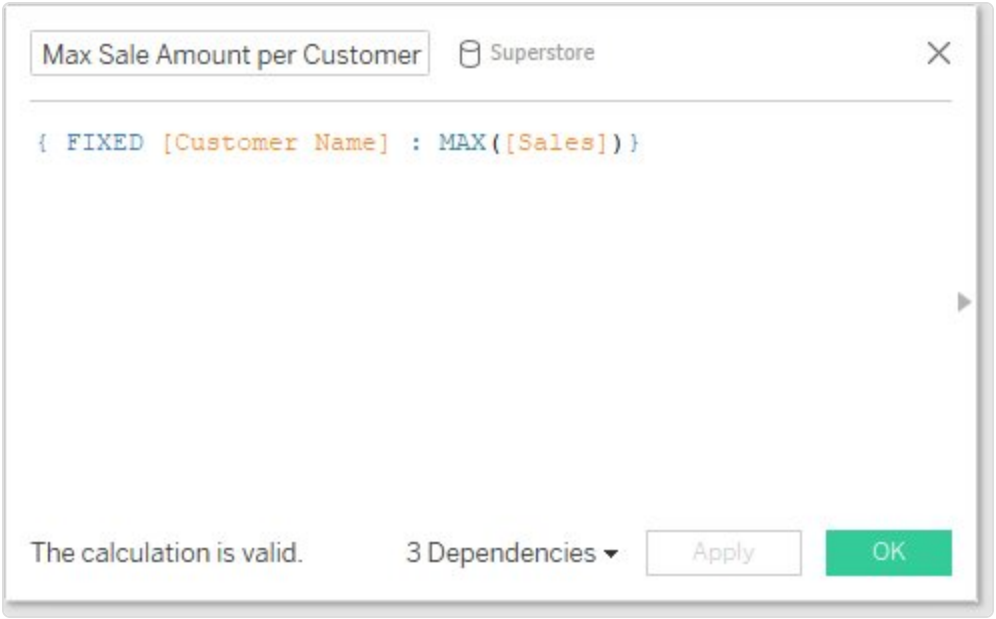
Exercise 1a

What is the gap between Customers' 1st and 2nd highest Sales value?



1. Write a calculation to find the highest Sales value per Customer.

   Calculation:

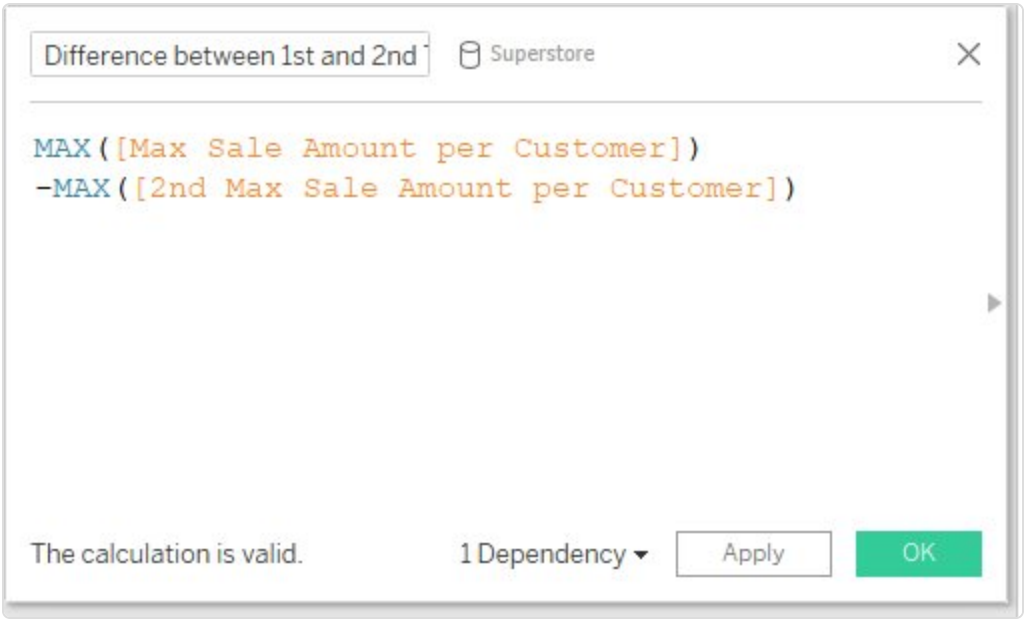


2. Write a calculation to find the 2nd highest Sales Amount per Customer.

   Calculation:

3. Write a calculation to find the difference between calculation 1 and 2.

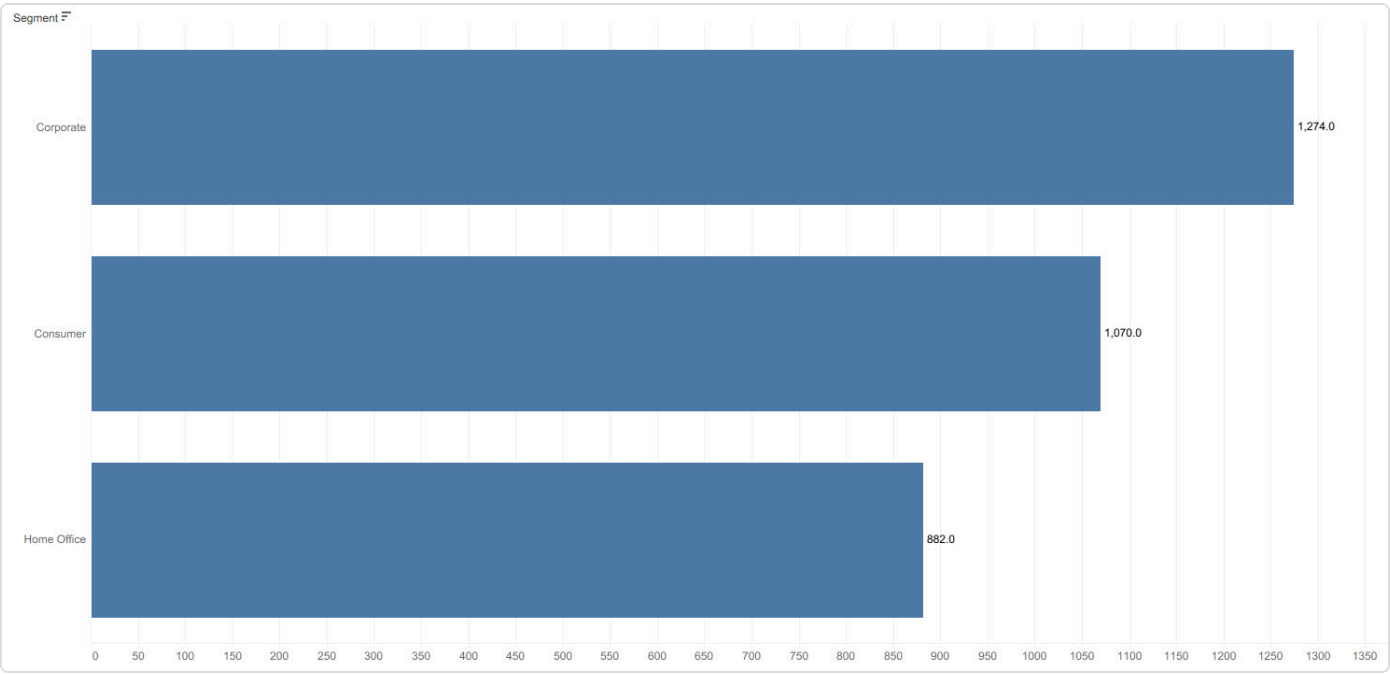
   Calculation:



4. Place each of these calculations on the Columns shelf.

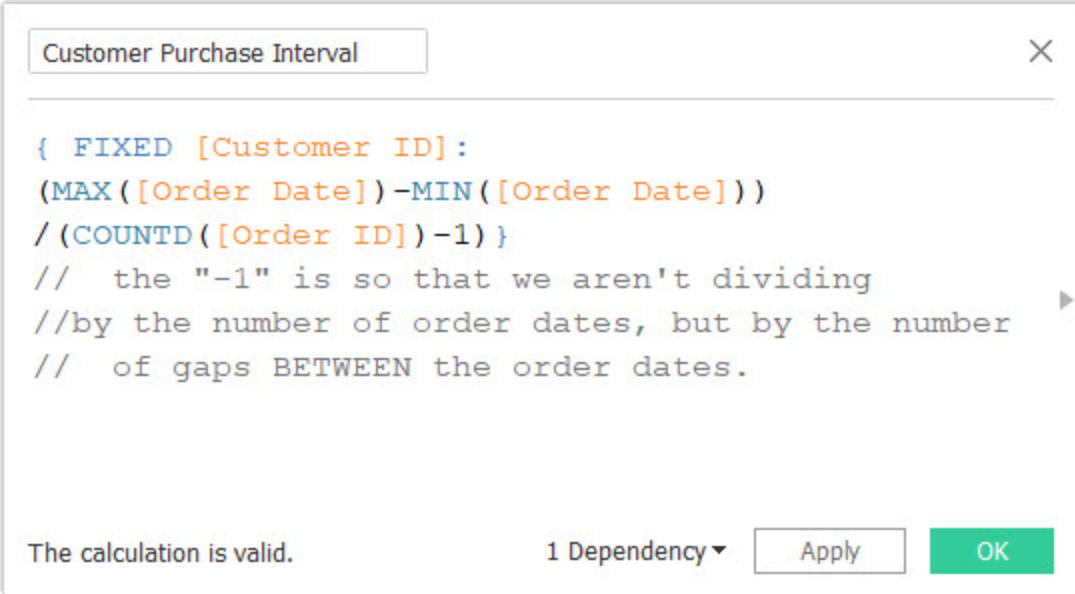
Challenge 1b

Find the average number of days between purchases for each Customer (Customer Purchase Interval). What is the highest Customer Purchase Interval in each Customer Segment?



1. Write a calculation that finds the average gap size between orders, per customer.

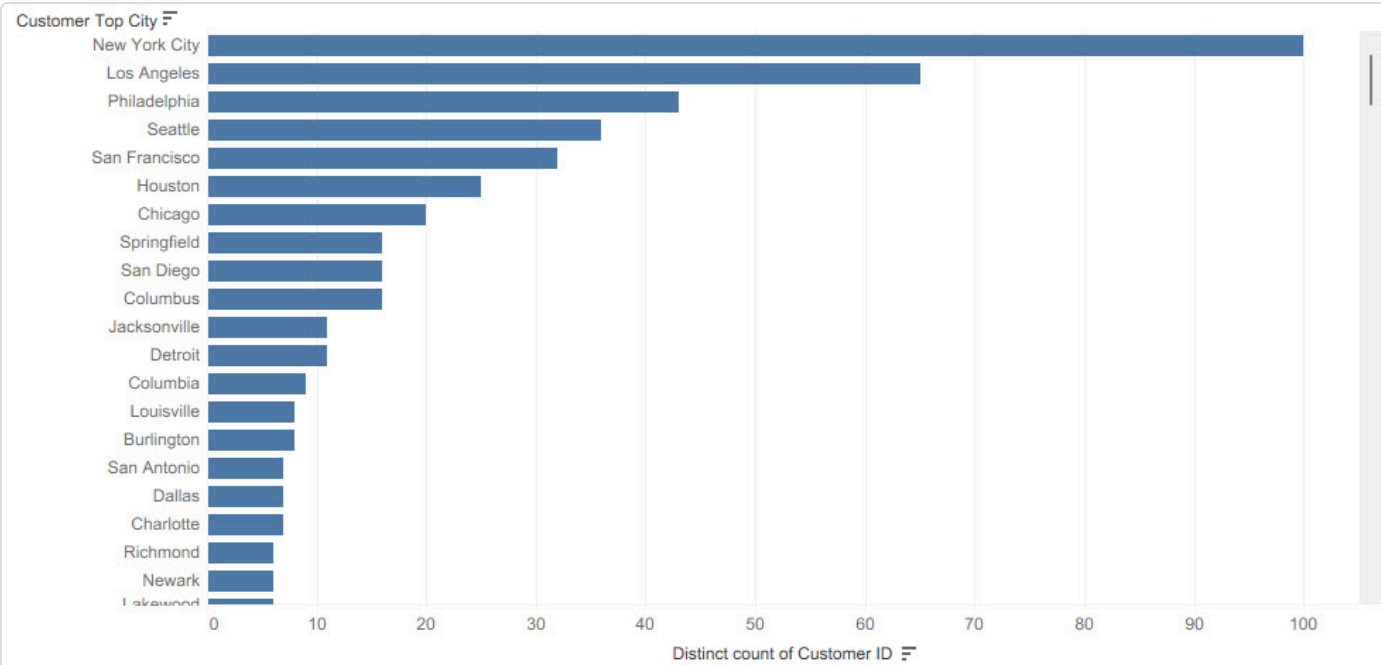
   Calculation:



2: Pivot, Isolate, Propagate

Exercise 2a

In which City did each Customer spend the most in one Sale?

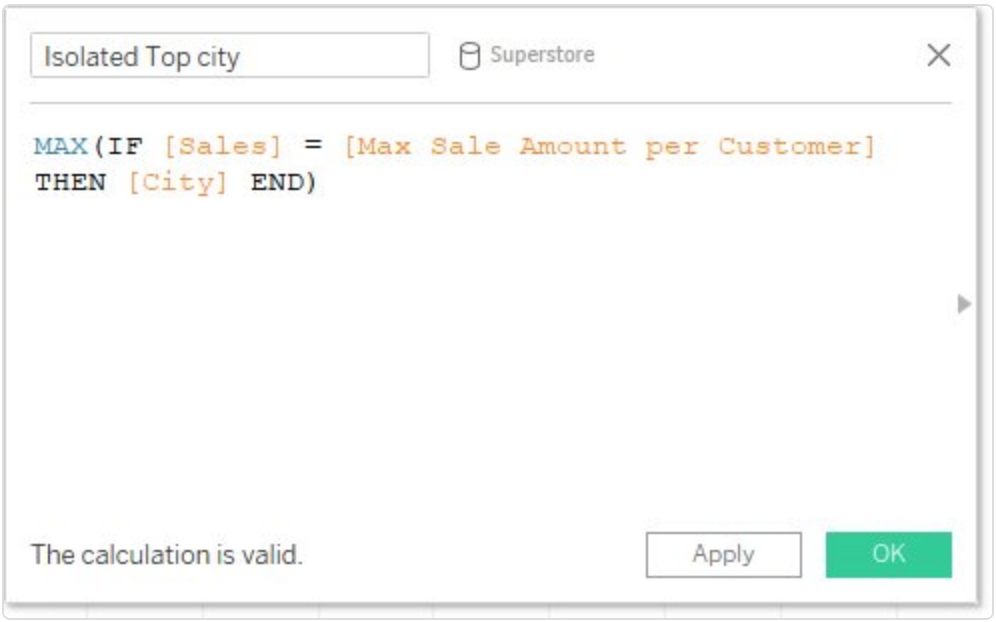


1. If not already created from challenge 1a, write a calculation to find the Max Sale Amount for each Customer.

   Calculation:

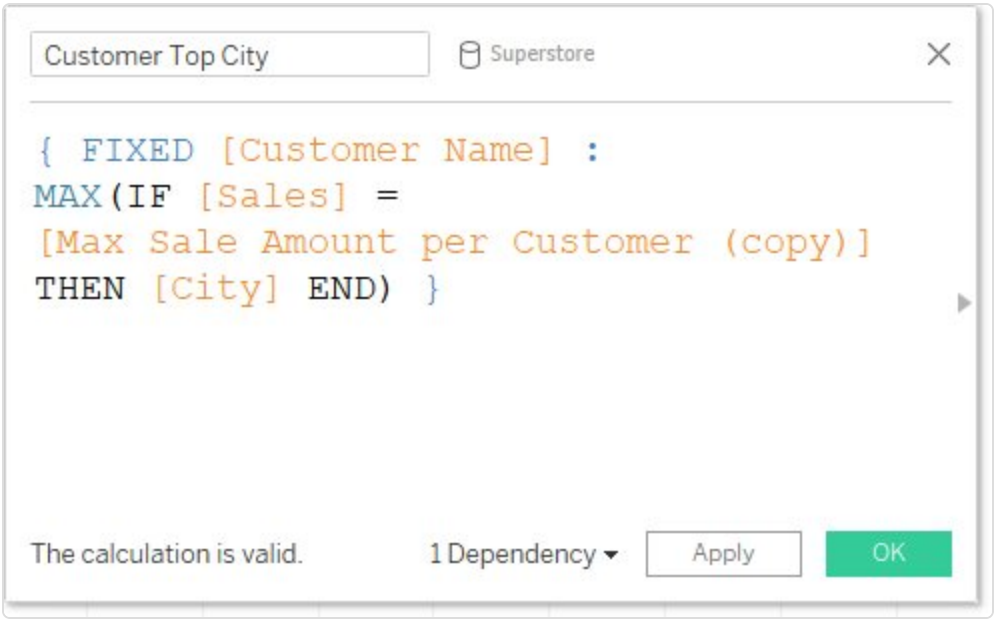
2. Write a calculation to isolate the Top City for each Customer (for simplicity, the calculation can be inserted directly into the next calculation if desired).

   Calculation:



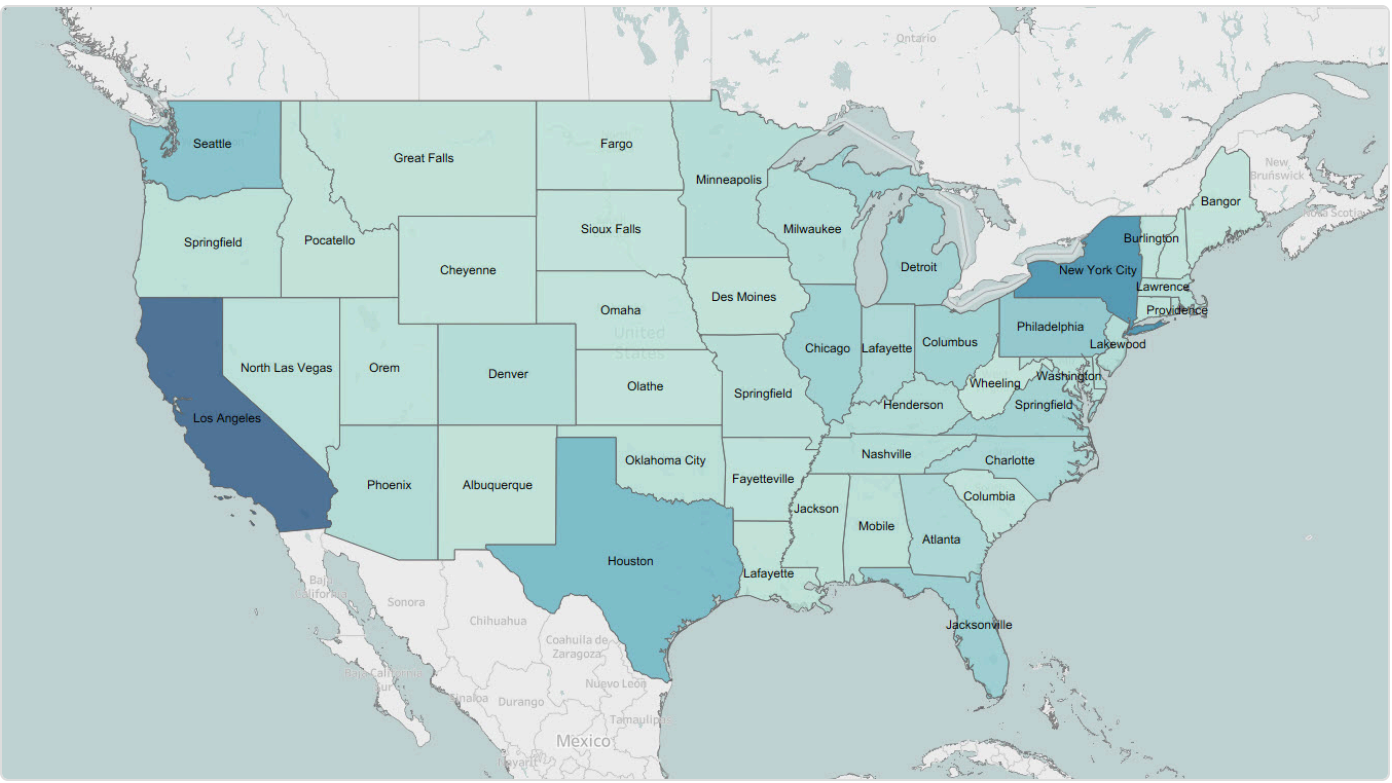
3. Write a calculation to propagate each Customer's Top City to all the rows for that Customer.

   Calculation:



Challenge 2b

View Sales by State on a filled map. Label each State with the City that has the highest Sales in that State.

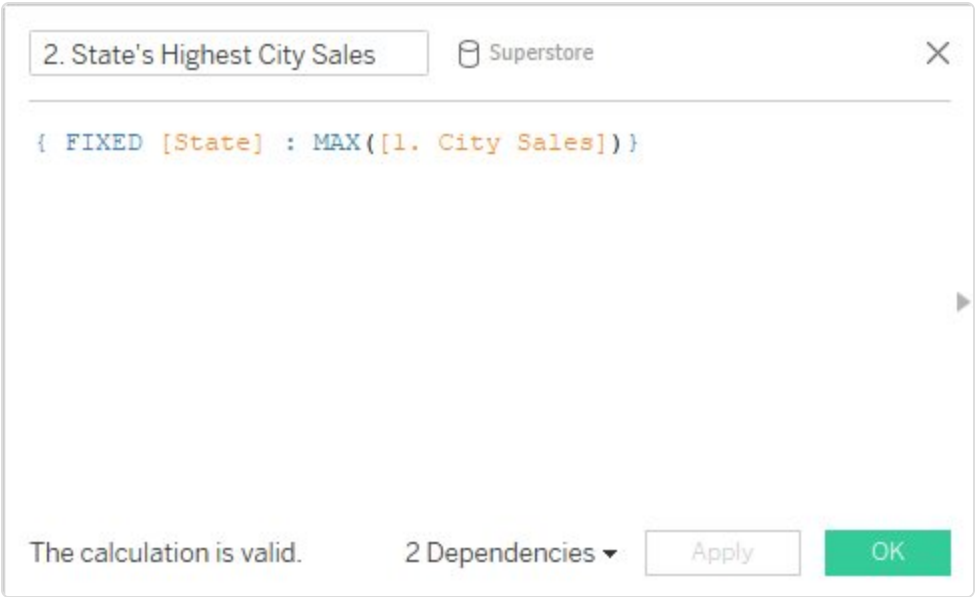


1. Write a calculation to find the total Sales for each City.

   Calculation:

2. Write a calculation to find the name of the City with the highest Sales for each State.

   Calculation:



3. Write a calculation to propagate the name of the Top City for each State across all rows of data.

   Calculation:

4. Add the final calculation to Text.

3: Selection Comparisons

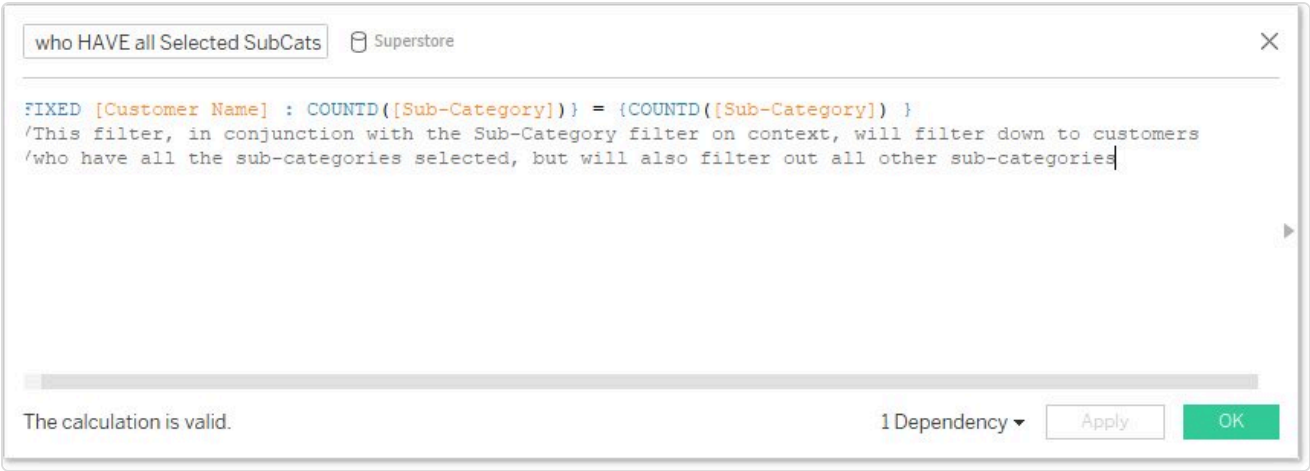
Exercise 3a

Show the Customers that have purchased from all the Sub-Categories selected in the filter. Select Fasteners, Machines, and Supplies.

Customer Name	Sub-Category: Fasteners	Machines	Supplies
Arthur Prichep	$41	$686	$7
Caroline Jumper	$12	$4,000	$3
Chloris Kastensmidt	$25	$1,070	$19
Christine Sundaresam	$5	$480	$28
Dean Katz	$17	$482	$9
Greg Tran	$46	$1,380	$42
Maxwell Schwartz	$4	$838	$311
Scott Cohen	$6	$240	$39
Sean Miller	$19	$23,460	$3
Stewart Visinsky	$6	$372	$25

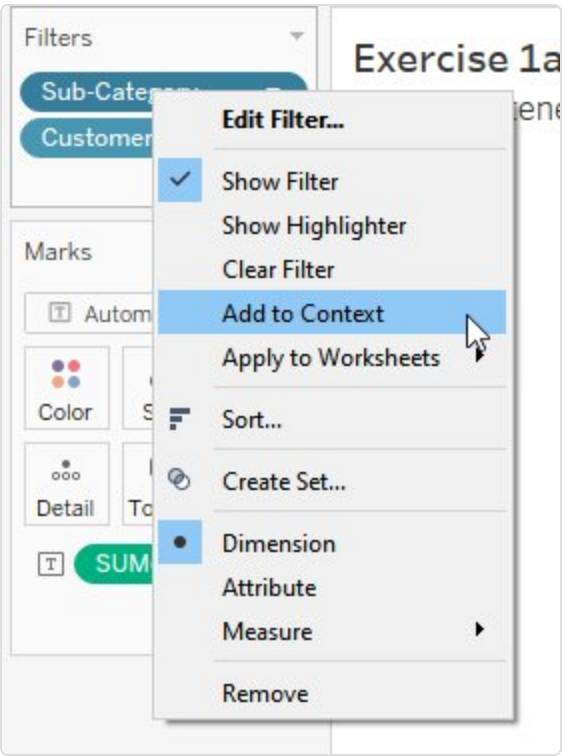
1. Write a calculation that compares a customer's # of Sub-Categories to the total # of Sub-Categories. Filter to TRUE.

   Calculation:



2. Right-click on the Sub-Category filter and select 'Add to Context'.

Screenshot:



Challenge 3b

Show all the data for Customers that have purchased from all the Sub-Categories selected in the Set list. Select Fasteners, Machines and Supplies.

	Sub-Category															
Customer Name	Faste..	Mach..	Suppl..	Acce..	Appli..	Art	Binde..	Book..	Chairs	Envel..	Furni..	Labels	Paper	Phon..	Stora..	Tables
Arthur Prichep	$41	$686	$7	$93	$100	$244	$154	$181	$113	$24	$233		$400	$144	$62	$843
Caroline Jumper	$12	$4,000	$3	$240	$395	$100	$103	$1,200	$4,058		$75	$44				$933
Chloris Kastens..	$25	$1,070	$19	$172	$58	$215	$227		$128	$90	$227		$165	$638	$118	
Christine Sunda..	$5	$480	$28	$1,100	$187	$73	$91	$242	$233	$15		$10	$376	$362		
Dean Katz	$17	$482	$9		$862	$3	$18		$770	$31	$3		$48	$24	$2,223	$313
Greg Tran	$46	$1,380	$42		$208	$69	$2,311	$4,008	$2,167	$51	$44		$654	$597	$243	
Maxwell Schwar..	$4	$838	$311	$280	$43		$98		$533	$136			$126	$46	$729	
Scott Cohen	$6	$240	$39	$70	$34	$7	$248	$1,553			$10		$75	$80	$714	$654
Sean Miller	$19	#####	$3	$22		$8	$100				$680		$89		$663	
Stewart Visinsky	$6	$372	$25	$28	$42	$79	$38		$2,044		$75		$148	$451	$434	$365

Selected Sub-Categories

- [ ] (All)
- [ ] Accessories
- [ ] Appliances
- [ ] Art
- [ ] Binders
- [ ] Bookcases
- [ ] Chairs
- [ ] Copiers
- [ ] Envelopes
- [x] Fasteners
- [ ] Furnishings
- [ ] Labels
- [x] Machines
- [ ] Paper
- [ ] Phones
- [ ] Storage
- [x] Supplies
- [ ] Tables

1. Create a calculation that compares a Customer's count of selected Sub-Categories to the total count of selected Sub-Categories. Add to the filter shelf.

   Calculation:

Customer who HAVE all Selected Superstore

```
{ FIXED [Customer ID] : COUNTD(IF [Selected Sub-Categories] THEN [Sub-Category] END)}
=
{ FIXED : COUNTD(IF [Selected Sub-Categories] THEN [Sub-Category] END)}

//This filter will filter down to the Customers with all the sub-categories selected, but
// it will retain all the data for those Customers
```

The calculation is valid. 1 Dependency Apply OK

Elite Challenge 1c: Using Measure Names & Values in a Calculation

Elite Challenge

Add conditional formatting to the crosstab:

- **Red Text:** Average Discounts in the Consumer or Home Office Segments that are >= 15%
- **Other Text:** Black

		Consumer	Corporate	Home Office
Furniture	Discount	19	18	16
	Profit	5,376	7,243	3,194
	Quantity	4,038	2,200	1,366
	Sales	372,634	203,771	108,998
Office Supplies	Discount	16	16	14
	Profit	49,420	34,862	26,787
	Quantity	10,534	6,247	3,897
	Sales	327,928	196,842	114,048
Technology	Discount	14	14	14
	Profit	70,447	40,646	35,796
	Quantity	3,430	1,988	1,231
	Sales	416,031	239,741	185,447

1. Write a calculation called **"Random Values"**. This calculation will give a random decimal value between 0 and 1, and must be computed at the row level of the data.

   Calculation:

2. Write a calculation called **"Field Names"**. This calculation will randomly assign a text value of "Discount", "Profit", "Quantity", or "Sales". Drag the field to the Rows shelf.

   Calculation:

Field Names

```
IF [Random Values] <.25 THEN 'Discount'
ELSEIF [Random Values] <.5 THEN 'Profit'
ELSEIF [Random Values] <.75 THEN 'Quantity'
ELSE 'Sales'
END
```

The calculation is valid. 4 Dependencies Apply OK

3. Write a calculation called **"Field Values"**. This calculation will leverage the Field Names calculation to reassign the correct values to the crosstab view. Drag this field to Text on the Marks Card.

   Calculation:

Field Values

```
IF [Field Names] = 'Discount' THEN {EXCLUDE [Field Names] : AVG([Discount])}*100
ELSEIF [Field Names] = 'Profit' THEN {EXCLUDE [Field Names] : SUM([Profit])}
ELSEIF [Field Names] = 'Quantity' THEN {EXCLUDE [Field Names]: SUM([Quantity])}
ELSEIF [Field Names] = 'Sales' THEN {EXCLUDE [Field Names] : SUM([Sales])}
END
```

The calculation is valid. 3 Dependencies Apply OK

4. Write a calculation called **"Formatting"**. This calculation will leverage the Field Names and Field Values calculations to apply the desired conditional formatting to the applicable cells of the crosstab. Drag this field to Color and Size on the Marks Card.

   Calculation:

Formatting

```
IF MIN([Field Names]) = "Discount" AND
    (MIN([Segment]) = "Consumer"
        OR MIN([Segment]) = "Home Office")
    AND MIN([Field Values]) >= 15 THEN "Red"
ELSE "Black"
END
```

The calculation is valid. 2 Dependencies Apply OK

Elite Challenge 2c: Top Sales Reps

Elite Challenge

Filter to the Sales Reps who won top seller for any single year. Display their total Sales for all time. Display their total Sales for the year they were top seller. Display their total Sales for the year they were a top Rep as a % of the Top Rep Amount for a selected Benchmark Year.

Benchmark Year (BY)

2020

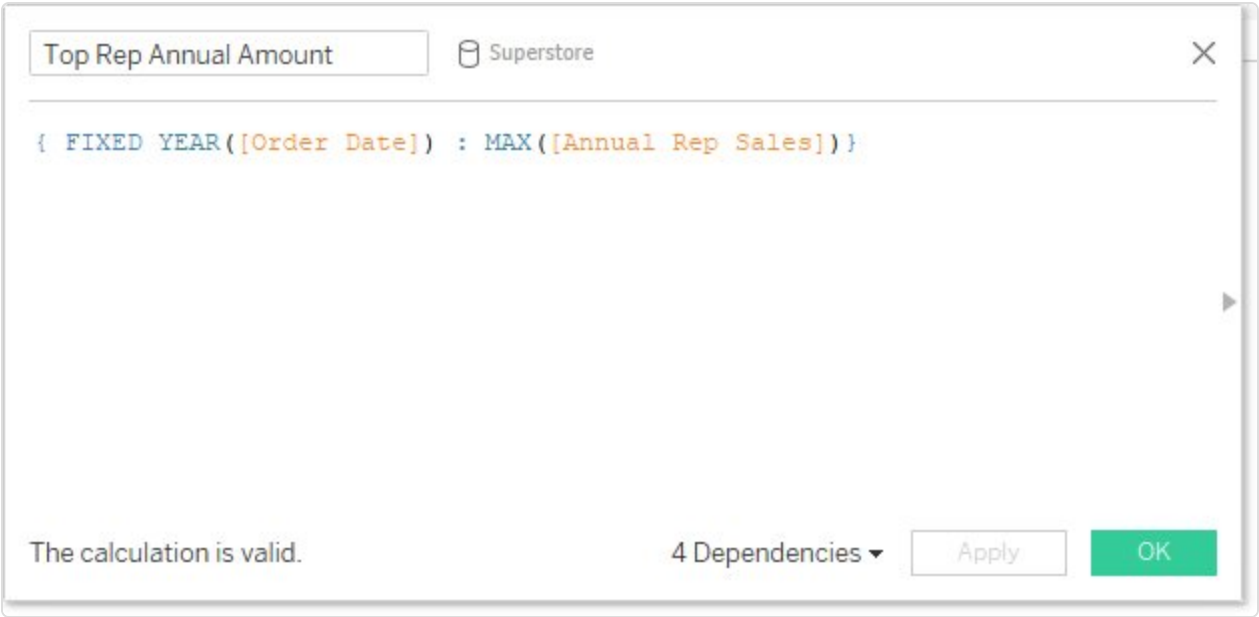
Sales Representative	Sales	Winning Amount	Winner Comparison Rate
Sean Miller	$25,043	$23,661	100%
Peter Fuller	$9,063	$9,022	38%
Tamara Chand	$19,052	$18,344	78%
Raymond Buch	$15,117	$14,203	60%
Patrick O'Donnell	$2,493	$467	2%

1. Write a calculation to find the Total Sales amount of each Rep, for each Year.

   Calculation:

2. Write a calculation to find the Top Rep's Sales amount, for each Year.

   Calculation:



3. Write a calculation to isolate the Winning Amount for the winner for each Year.

   Calculation:

4. Write a calculation to filter to the Reps who's annual amount equals the Top Rep Annual Amount. Add this as a filter. You should now see five Reps in the view.

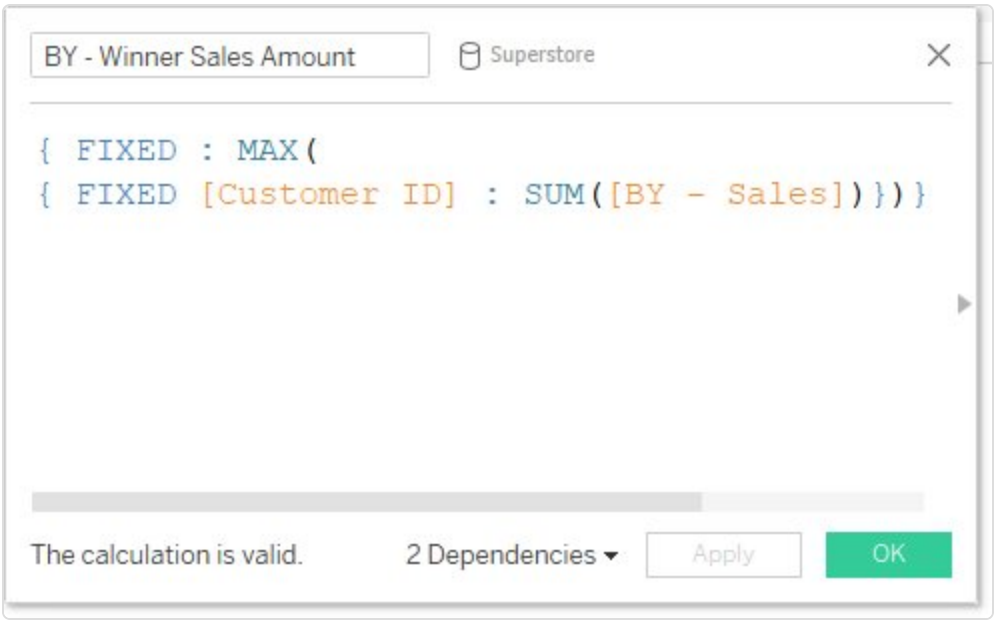
   Calculation:

5. Write a calculation to filter to the Benchmark Year of Sales.

   Calculation:

6. Write a calculation to find the total amount of Sales for the Rep with the highest Sales in the Benchmark Year.

   Calculation:



7. Write a calculation to find the ratio between the Winning Amount for the Sales Rep overall, and the Winning Amount of the Benchmark Year Winner.

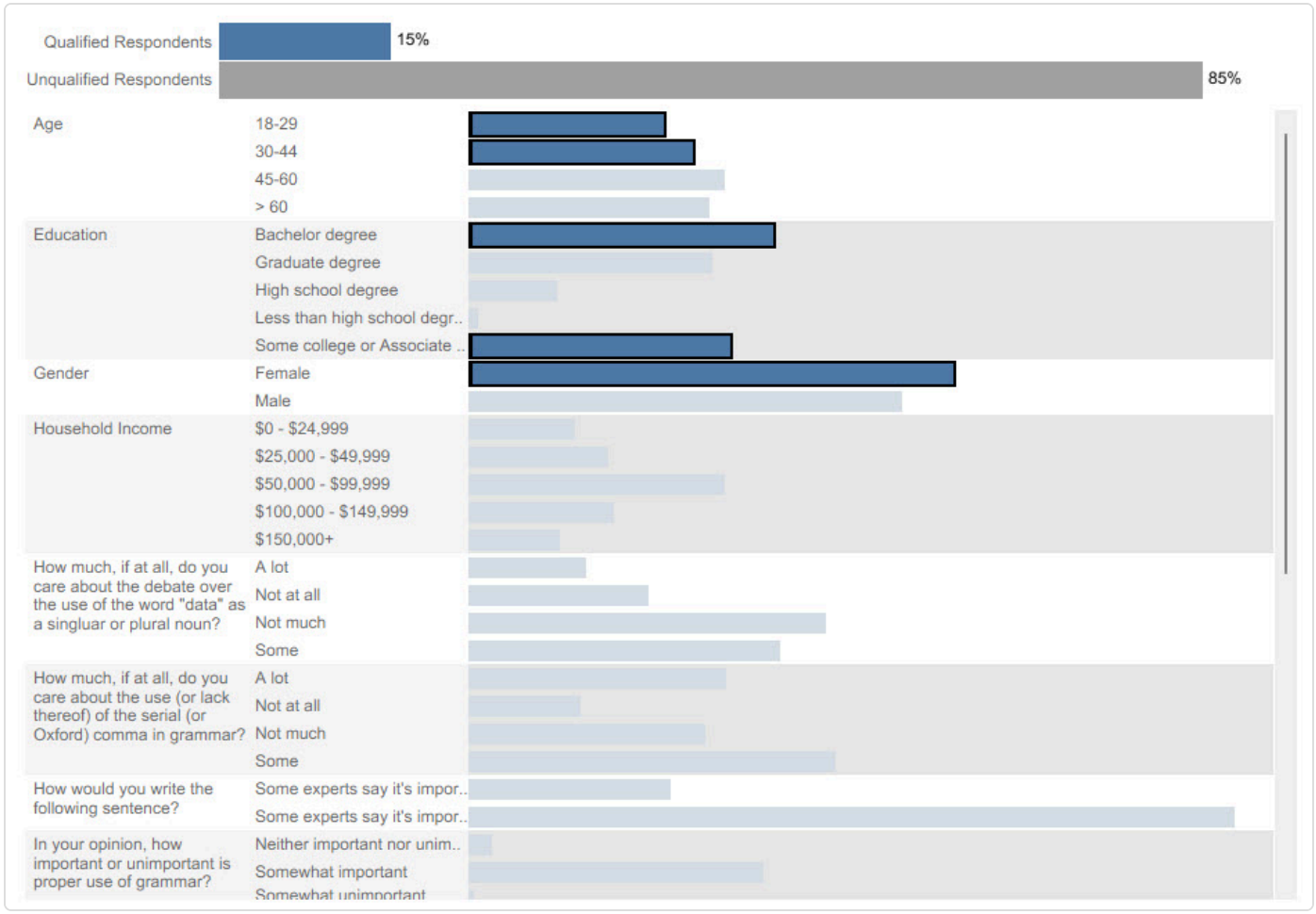
   Calculation:

8. Add the Winning Amount and the Winner Comparison Rate to the view.

Elite Challenge 3c: Surveys and Sets

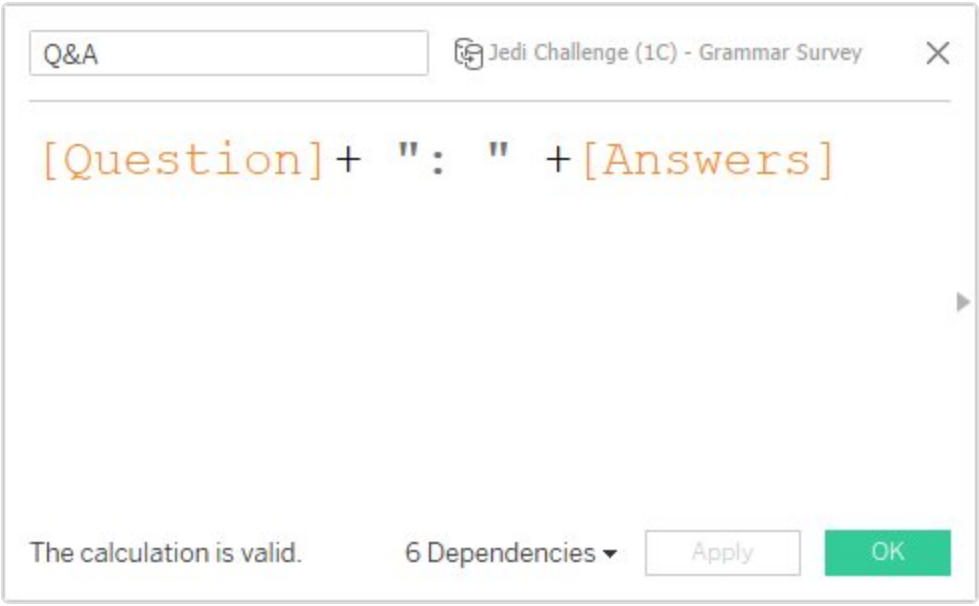
Elite Challenge

View the % of Respondents who answered each Question with an Answer selected.



1. Create a calculation combining the Question and Answer fields.

   Calculation:

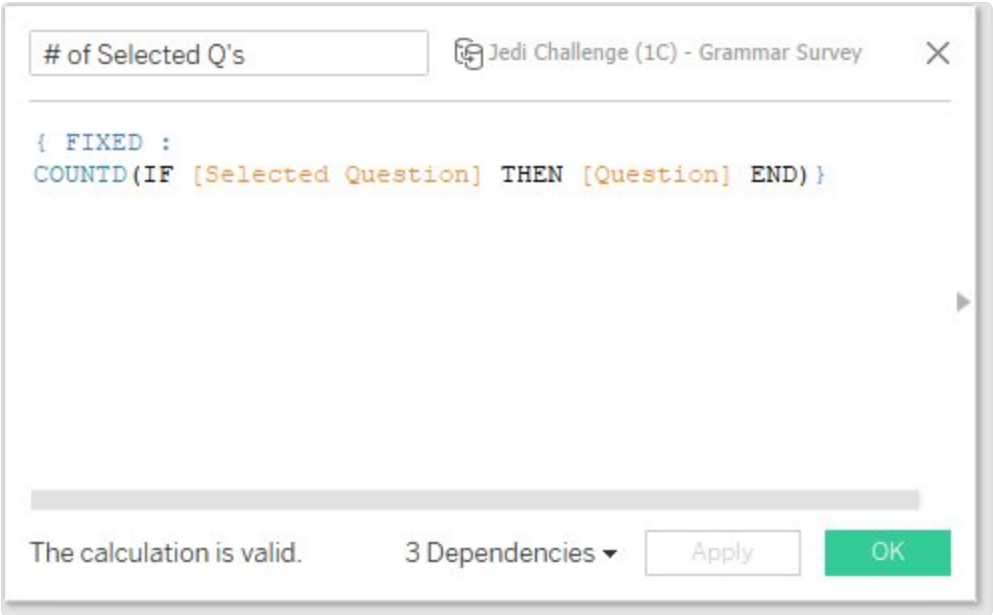


2. Create a Set on Question, called **"Selected Question"**. (Right-click on Question, and select "Create > Set")

3. Create a Set on Q&A, called **"Selected Q&A"**. (Right-click on Q&A, and select "Create > Set")

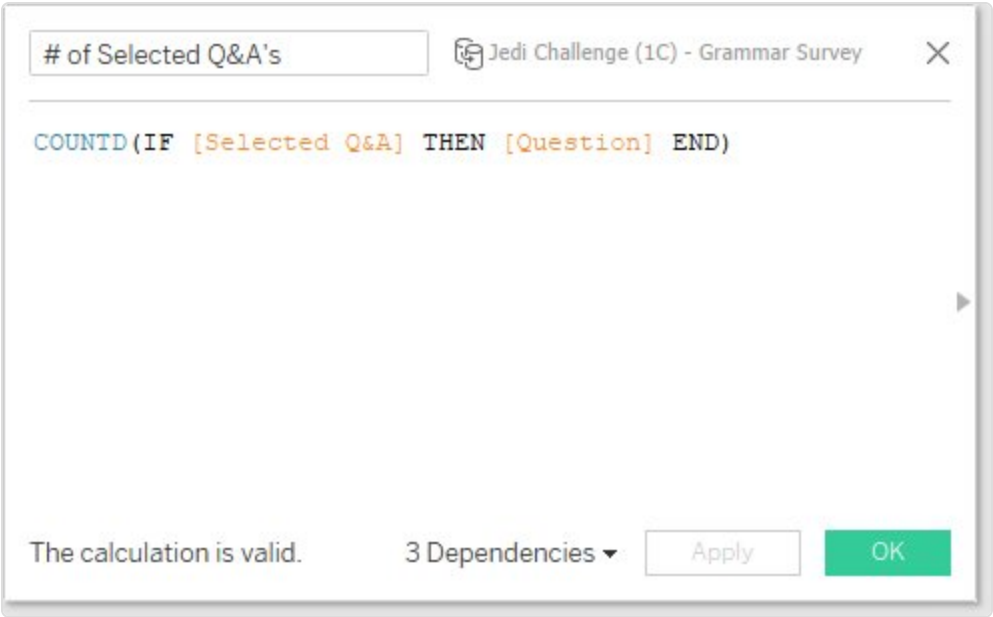
4. Write a calculation for the number of selected Questions.

   Calculation:



5. Write a calculation for the number of selected Q&A's.

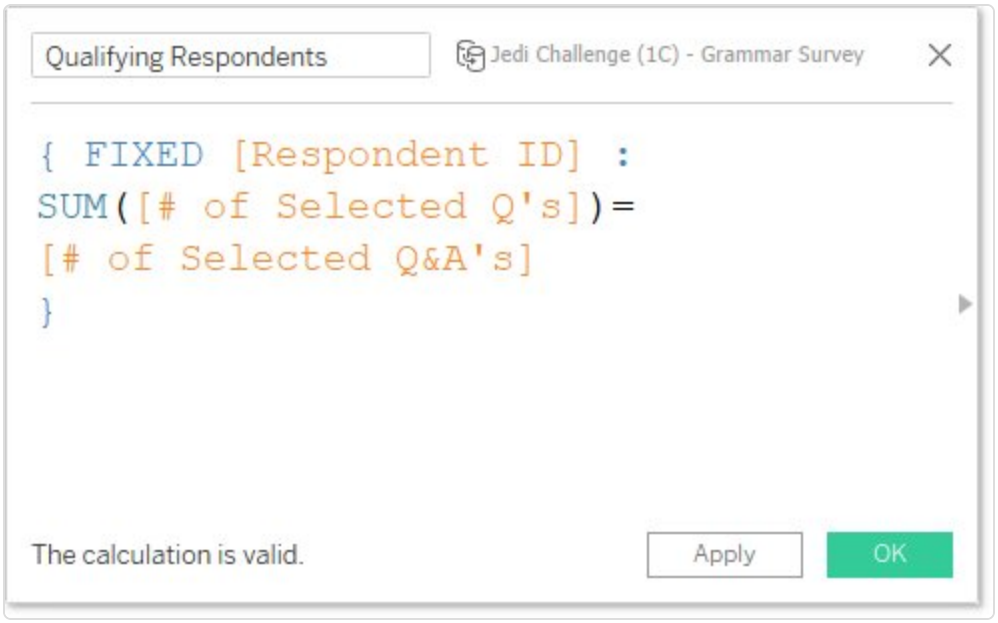
   Calculation:



6. Write a calculation to find the respondents that have the same amount of Q&A's selected as total Questions selected. Or, create a Conditional Set to find the same result:

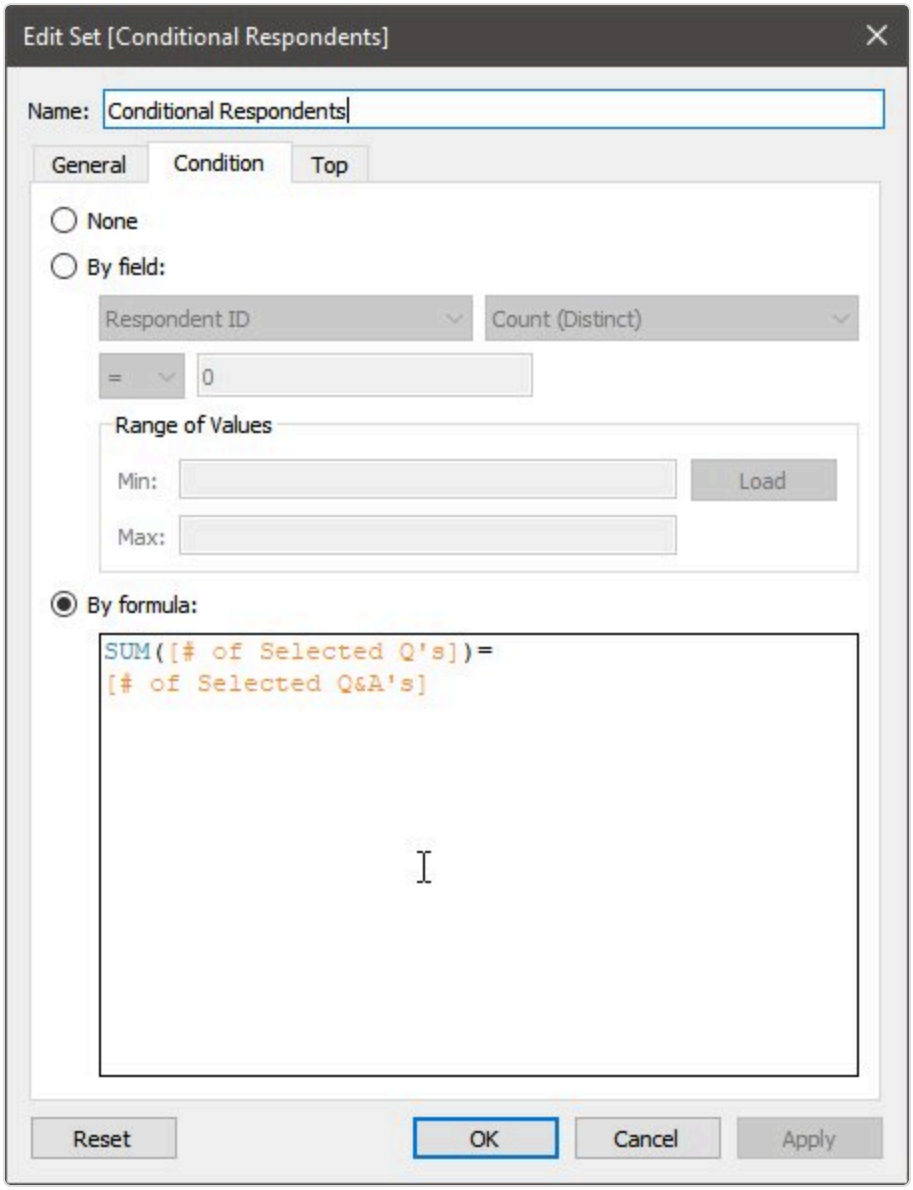
   A. Calculation Option:

   Calculation:



   B. Set Option:

   Edit Set dialog:

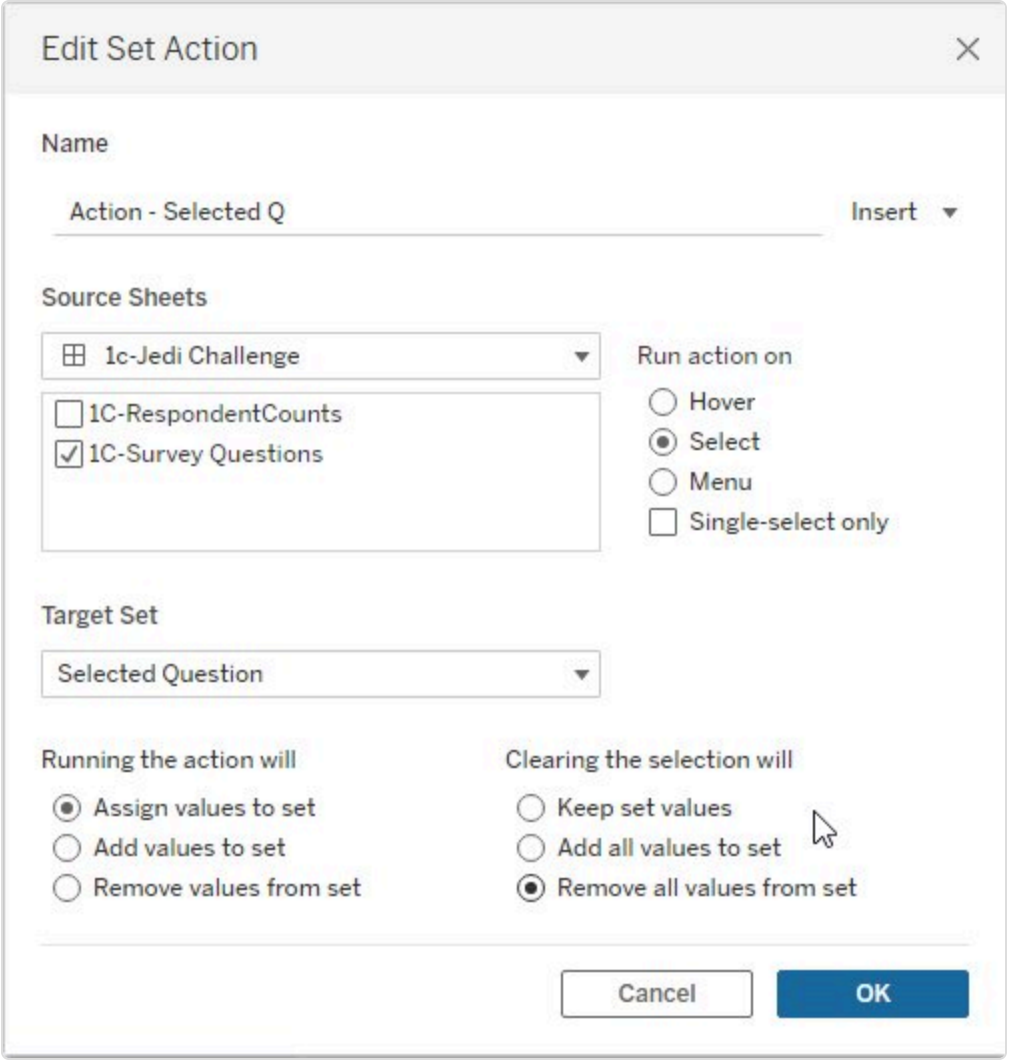


7. Add the calculation or set to the row's shelf on the **"1C-RespondentCounts"** sheet. Edit Aliases of True/False values if desired.

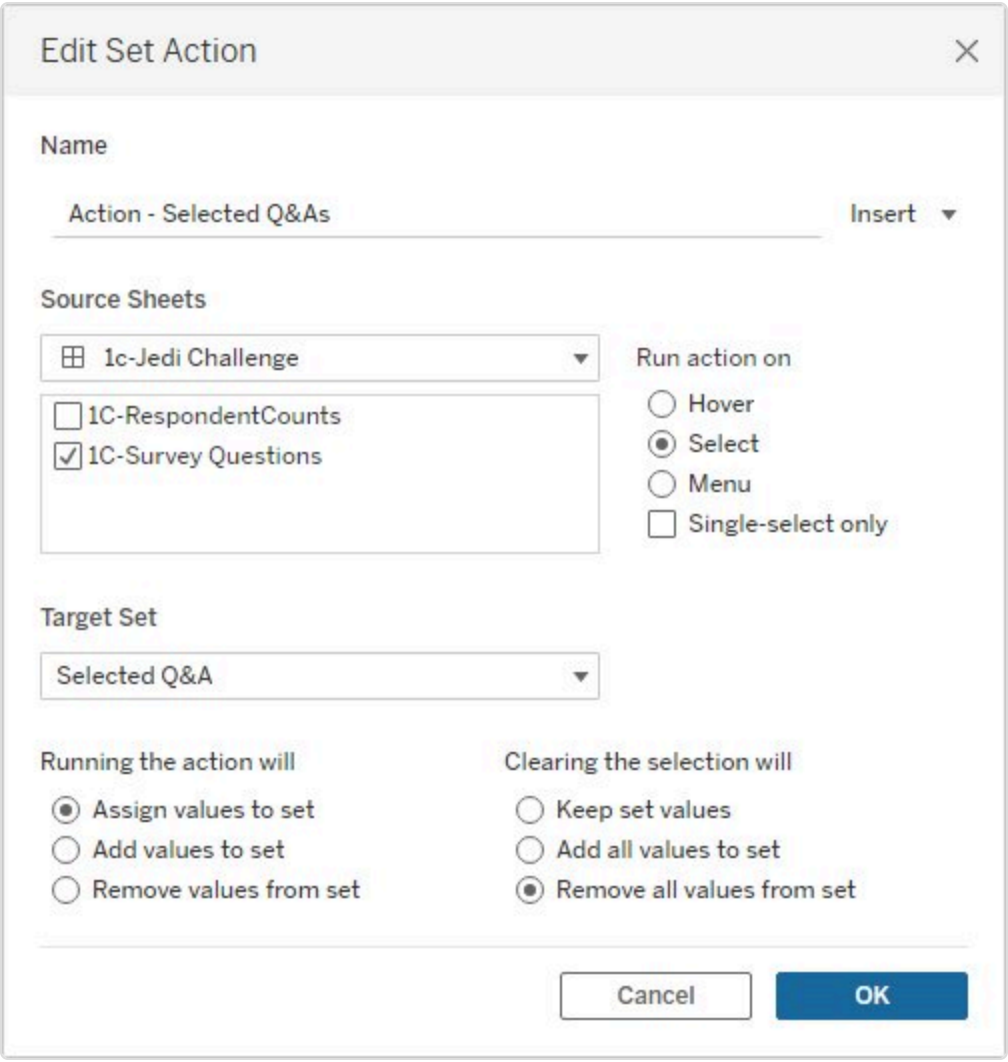
   Screenshot:

8. Return to the dashboard for 1C. (Create 2 Set Actions by going to **Dashboard > Actions > Add Action > Change Set Values**)

   A. Set Action #1: Selected Questions:



B. Set Action #2: Selected Q&A's:



> Hold the Ctrl (or Cmd key for Mac) and select multiple bars in the bottom view of the dashboard. The Qualified Respondents % should update accordingly.